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PAPER 69-43

PALEOMAGNETISM OF THE LATE MIOCENE PLATEAU BASALTS IN THE CARIBOO REGION OF BRITISH COLUMBIA

(Report and 4 figures)

D. T. A. Symons

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CONTENTS

	Page
Abstract	v
Introduction	1
Sampling	2
Remanence measurement and demagnetization	3
Remanence data selection	3
Reliability and stability of remanence measurements	3
Analysis and discussion of remanence directions	5
1. Intrasectional reversals	5
2. Division of flows by polarity	5
3. Division of flows by section	6
4. Division of flows by stratigraphy	7
5. Undivided result	7
6. Comparison with other paleomagnetic results	9
Conclusions	9
References	10
Appendix – Tables 1-3	13
Table 1. Site remanence data	14
2. Summary of remanence data	15
3. Pole positions	16

Illustrations

Figure 1. Site location map	2
2. Histograms	4
3. Schmidt equal-area plots showing mean remanence directions	6
4. Pole positions shown in the northern hemisphere north of latitude 30°N	8

ABSTRACT

A paleomagnetic study was made on 251 cores from 49 flows representing several stratigraphic sections of undisturbed olivine-rich plateau basalts in the Cariboo region of south-central British Columbia. Alternating-field demagnetization was used to isolate the primary thermomagnetic remanence of the cores. Statistical analysis of the remanence directions indicates that the basalts were extruded over a time interval of between 2 and 5 m.y., and that they emanated from numerous volcanic vents in local accumulations which formed rapidly within one polarity interval. The derived pole position of 140.3°W, 84.0°N (δp 4.8°, δm 5.4°) is consistent with: (1) the late Miocene (14 to 10 m.y. ago) age indicated by radiometric and paleontologic evidence; (2) the pole derived from four gabbroic plugs which apparently occupy the original volcanic vents; and (3) the pole derived from the Columbia River basalts which indicates that the flows have not been subjected to regional tectonic rotation.

PALEOMAGNETISM OF THE LATE MIOCENE PLATEAU BASALTS IN THE CARIBOO REGION OF BRITISH COLUMBIA

INTRODUCTION

Plateau basalts are exposed throughout a northwest-southeast rectangular area of about 300 miles in length from Kamloops Lake in the south to Babine Lake in the north, and about 100 miles in width from the Coast Mountains on the west to the Cariboo Mountains on the east. They cover approximately two-thirds of this area forming much of the broad flat Interior Plateau between prominent hills of older rocks. In the southern portion of this area which is the region of this study (Fig. 1) they are simply called the plateau basalts. Despite their considerable areal extent, relatively little geological information is available about them.

The plateau basalts are composed of dark vesicular olivine basalt in thin prismatically-jointed lava sheets with minor local brecciation and pillow structure. Foreset bedding can occasionally be seen. Mathews and Rouse (1963) reported minor amounts of diatomite, shale and unconsolidated sandstone associated with them. The plateau basalts are flat lying with dips of less than 1 degree except in rare local cases such as slump blocks. They are found between the 2,500- and 4,000-foot elevation contours except on the western flank where the unit rises to 8,500 feet and caps some of the Coast Mountains as the result of post-extrusive elevation of the Coast Mountains and subsequent dissection of the plateau (Tipper, 1963; Campbell and Tipper, 1966). The thickness of this unit is highly variable but rarely exceeds a few hundred feet with maximum-thickness estimates ranging up to 1,000 feet (Tipper, 1959; Muller and Tipper, 1962).

The plateau basalts overlie units of Miocene age conformably and a variety of older units with angular unconformity (Tipper, 1957 and 1960; Campbell, 1961). They are overlain by Pleistocene glacial deposits except for a few Recent lava flows and cinder cones (Campbell, 1961 and 1963; Tipper, 1959). Mathews and Rouse (1963) placed the age of the plateau basalts as late Miocene to early Pliocene on the basis of paleontologic evidence derived from fossils in the sediments and on three K-Ar radiometric dates of 13 ± 2, 12 ± 2 and 10 ± 2 m.y. on ash horizons at the base or in the lower portion of the lava sequence. Their three samples came respectively from near the Beaumont Creek, Deadman Falls and Hanceville sections sampled in this study (Fig. 1). They assigned the geological equivalent age using the Holmes'

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Project No.: 680052.
Author's address: Geological Survey of Canada,
601 Booth Street,
Ottawa 4, Canada.

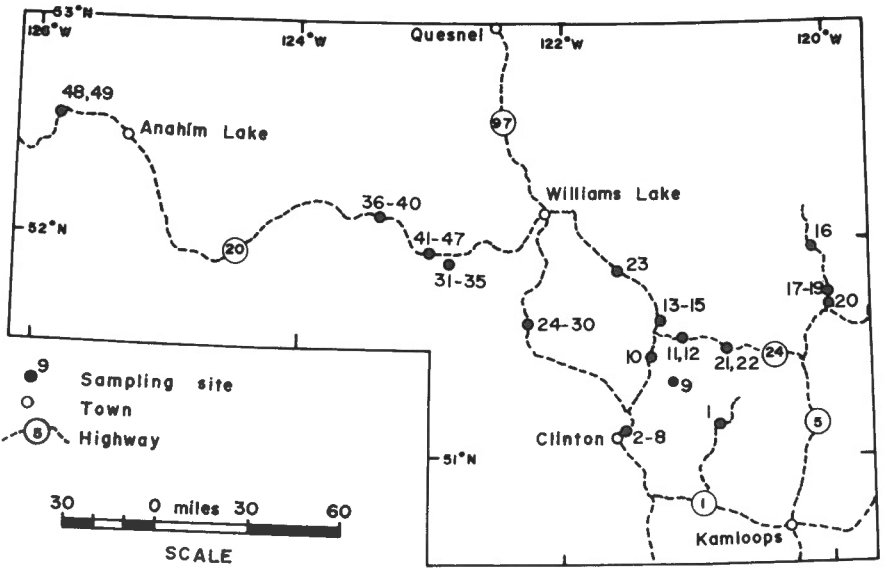


Figure 1. Site location map. The sections are Deadman Falls (site 1), Clinton Chasm (2-8), Tin Cup Mountain (9), Mount Begbie (10), Forestry Hill (11, 12), 100 Mile House (13-15), Clearwater Lake Road (16-20), Bridge Lake (21-22), Lac la Hache (23), Dog Creek (24-30), Beaumont Creek (31-35), Bull Canyon (36-40), Hanceville (41-47), and Heckman Pass (48-49).

(1959) time scale (Mathews, 1963). Using the Geological Society of London (1964) time scale, the geological equivalent age for the plateau basalts would be late Miocene.

The source vents for these basalts have not been located except perhaps in the extreme south where four olivine gabbro plugs intrude the flows (Farquharson, 1963).

No previous paleomagnetic work on the plateau basalts has been reported. The purpose of this study was to determine their paleomagnetic properties and thereby: (1) to establish a reliable pole position for the late Miocene; (2) to examine the possibility of correlation between the sections; (3) to compare them to the results found for the 'source-vent' gabbroic plugs (Symons, 1969); and (4) to compare them to the results found by Watkins (1965) for the apparently contemporaneous Columbia River basalts and other plateau lavas to the south in the states of Washington, Idaho and Oregon, U.S.A.

SAMPLING

There are very few good stratigraphic sections of the plateau lavas exposed and, of these, most are too steep to be sampled. The locations of the sampled sections are shown in Figure 1. A total of 251 cores were collected from 49 individual flows (sites). Generally, 4 to 6 cores were collected from each site (Appendix, Table 1) with individual cores taken from points

distributed across the vertical exposure of the flow and several feet apart. The cores were oriented in situ using a solar compass with Brunton compass and topographic orientation checks. Two cylinders each 1 1/4 inches in diameter and 1 3/32 inches in length were cut from each core.

REMANENCE MEASUREMENT AND DEMAGNETIZATION

Using an automated biostatic magnetometer (Larochelle and Christie, 1967), the direction and intensity of the natural remanent magnetization (NRM) was measured for each cylinder. One test cylinder from each site with a representative NRM was subjected to alternating-field (AF) demagnetization using the apparatus described by Larochelle and Black (1965) at successive peak field intensities of 25, 50, 75, 100, 200, 300, 400, and 800 oersteds, and its residual remanence measured after each step. Using the demagnetization data, the stability index (SI) (Tarling and Symons, 1967) was calculated for each of the 49 test cylinders (Appendix, Table 1). Based on the examination of the successive change in remanence direction and intensity during demagnetization and on the field intensity limits defined by the SI, a field of 200, 300 or 400 oersteds was selected for each site in which the remaining cylinders were demagnetized and their residual remanence measured. The remaining cylinders from sites 1, 2, 10, 15, 26, 35, 36, 48 and 49 were demagnetized a second time in a higher peak field of 400 or 800 oersteds, and for these sites the remanence data from the better grouping of directions is reported in Table 1 (Appendix).

REMANENCE DATA SELECTION

Two criteria were preset in order to assure that the remanence data used for statistical analysis were homogeneous and free from gross errors. Cores not meeting these criteria were rejected.

First, the remanence of a core was considered satisfactory if the angle θ between the remanence directions of the two cylinders from the core did not exceed 15 degrees. The vectorial average of the remanence directions from the two cylinders was used to represent the remanence direction of the core.

Second, the angular standard deviation for each site (δ_s) (Larochelle, 1968) was calculated giving each core unit weight. If the angular deviation of one core from the mean remanence direction of the remaining cores exceeded ($3 \times \delta_s$) for the remaining cores, then the one core was considered an erratic and rejected.

RELIABILITY AND STABILITY OF REMANENCE MEASUREMENTS

Several points of evidence suggest that orientation errors in collecting and measuring are minimal and that this suite of cores as a whole have a high degree of stability.

(1) Only 4 cores, or less than 2 per cent of the collection, failed to meet the first of the preset data selection criteria. Their failure could be due to

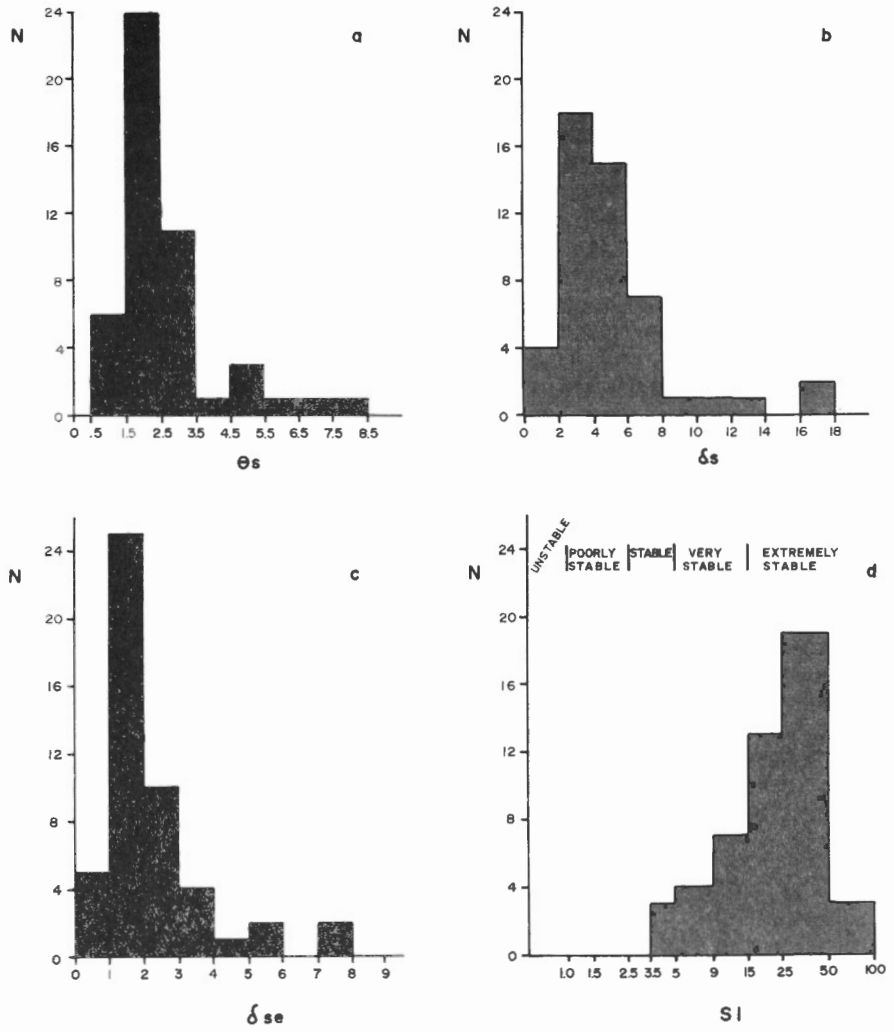


Figure 2. Histograms of the: a) site mean angles (θs) between the remanence vectors of specimens within each core; b) site mean angular standard deviations (δ_s); c) site mean angular standard errors (δ_{se}); and d) site stability index (SI) values.

either a gross orientation error in measurement, magnetization heterogeneity, or significant remanence instability.

(2) Only 10 cores or 4 per cent of the collection failed to meet the second of the preset data selection criteria. Their failure could be due to a gross orientation error in collecting, to a geologic reason such as reheating of the core by an adjacent flow, to a physical reason such as a lightning strike, or to a hard viscous component of remanence so that demagnetization did not leave a single primary component.

(3) The site mean value (θ_s) of the θ angles of accepted cores was calculated for each site (Appendix, Table 1), and the results are tabulated in Figure 2a. Values of θ_s of less than 1.5 degrees could be due to measurement orientation errors, but values more than 1.5 degrees are probably due to remanence instability or inhomogeneity of magnetization of the core. For this suite of cores θ_s of less than 2.5 degrees for about 60 per cent of the accepted cores and less than 6.0 degrees for 96 per cent which indicates a high degree of remanence directional stability and homogeneity.

(4) The site mean angular standard deviations (δ_s) are tabulated in Figure 2b. Values of δ_s of less than 2.0 degrees could be due to collection orientation errors, but values more than 2.0 degrees are probably due to inhomogeneity of magnetization of the site or the presence of minor viscous components of remanence that have not been removed by demagnetization. Excluding site 2 which has a transitional remanence direction and for which δ_s = 17.1 degrees, the value of δ_s is less than 4.4 degrees for 50 per cent of the sites and less than 9.0 degrees for 94 per cent of the sites which indicates a high degree of internal site consistency or remanence homogeneity. This stability and reliability is further shown by the low values of the angular standard error δ_{se} associated with each site (Fig. 2c).

(5) The SI values indicate the presence of a high degree of remanence stability in the test cylinders with 3 values falling in the 'stable' (2.5-5.0), 11 in the 'very stable' (5.0-15.0), and 35 in the 'extremely stable' classification of Tarling and Symons (1967) (Fig. 2d).

ANALYSIS AND DISCUSSION OF REMANENCE DIRECTIONS

The remanence direction results of this study will be examined from the following points of view: (1) intrasectional polarity reversals; (2) division of flows by polarity; (3) division of flows by section; (4) division of flows by stratigraphic elevation; (5) undivided result; and (6) comparison with results from other studies.

1. Intrasectional Reversals

Examination of the site mean remanence directions (Appendix, Table 2) reveals that there are no reversals within any of the stratigraphic sections so that it is not possible to define detailed paleomagnetic stratigraphy on the basis of reversals. There is perhaps one exception in the Clinton Chasm section (site 2) in which the lowermost sampled flow gives a stable transitional or intermediate remanence direction, that is the inclination is less than 50 degrees. In the statistical analyses and discussion that follow, this flow is excluded.

2. Division of Flows by Polarity

The flows were divided into two groups – those with normally polarized remanence directions and those with reversely polarized directions. Giving each site mean remanence direction unit weight, the mean declination (D_g) and inclination (I_g) was calculated for each group along with the cone of

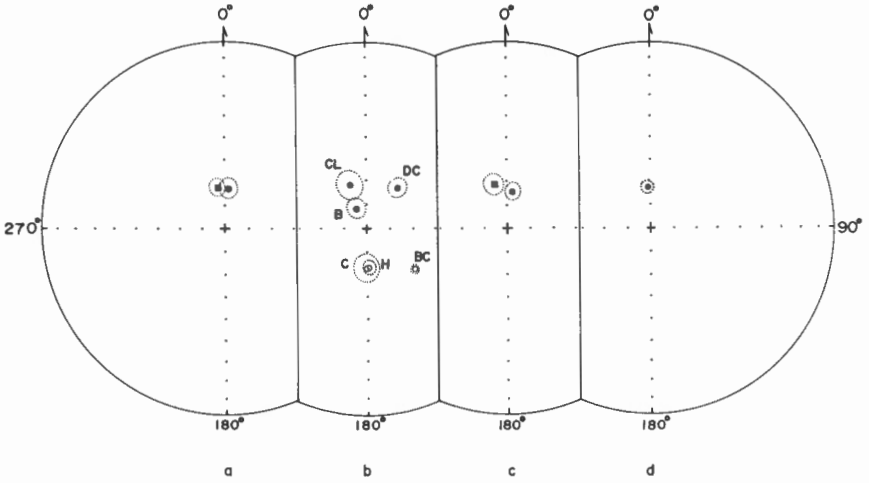


Figure 3. Schmidt equal-area plots showing the mean remanence directions from the: a) normally-polarized flows (circle) and normalized reversely-polarized flows (square); b) stratigraphic sections at Clinton Chasm (C), Clearwater Lake Road (CL), Dog Creek (D), Beaumont Creek (B), Bull Canyon (BC), and Hanceville (H); c) normalized lower (circle) and upper (square) stratigraphic flows; and d) all flows. Solid (open) symbols indicate positive (negative) or downwardly-directed (upwardly-directed) vectors. The dotted lines indicate the cone of 95 per cent confidence.

95 per cent confidence (α_{95}), angular standard deviation (δ_g) and angular standard error (δ_{ge}) (Larochelle, 1969) (Appendix, Table 2). Both groups have very similar mean remanence directions and distributions. Application of the angular variance ratio test (Larochelle, 1968) gives the statistic $F = 1.84$ which is much less than the theoretical value $F(2, 92, .05) = 3.10$ so that the normalized remanence directions of the normal and reverse flows (Fig. 3a) are not statistically significantly different other than in sign and hence the derived pole positions are not significantly different. The geological implication is that the flows were deposited during several normal and reverse polarity intervals representing the same time span.

3. Division of Flows by Section

The next step was to consider each stratigraphic section as an entity. Giving each normalized site mean remanence direction unit weight, the section mean declinations (D_x), etc., were calculated (Appendix, Table 3). The resulting section mean directions for the six sections composed of five or more flows are shown on Figure 3b. None of the sections are obviously anomalous or obviously correlative except perhaps the Clinton Chasm and Hanceville sections. Further, the small values obtained for α_{95} and δ_{xe} indicate that each of the six sections was likely deposited during one polarity interval. Cox (1968) has estimated the mean length of the polarity intervals

during the last 10.6 m.y. to be 0.22 m.y. based on known and postulated rates of polarity reversal of the Earth's magnetic field from radiometrically dated lava flows and sea floor spreading rates. Examination of the sea-floor magnetic patterns (Heirtzler, 1968) suggest that the value estimated by Cox would be approximately correct for the late Miocene. It is consistent also with the minimum of 22 reversals for the time interval 4 to 12 m.y. found in ocean floor sediments (Foster and Opdyke, 1969). Thus it appears that local accumulations of flows were rapidly deposited within one polarity interval, and that during several polarity intervals these accumulations abutted or interfingered upon deposition to form the present extensive plateau.

4. Division of Flows by Stratigraphy

Using the available geological maps, it is possible to assign each section to its relative stratigraphic position in one of five categories – lower (L), lower to middle (L-M), middle (M), middle to upper (M-U), and upper (U) (Appendix, Table 2). The flows were then combined into lower (L, L-M, M) and upper (M-U, U) stratigraphic groups using the normalized site mean remanence directions (Appendix, Table 2). It was found that the two groups have different remanence directions with similar distributions (Fig. 3c). Comparing the two populations using the variance ratio test gives the statistic $F = 8.55$ which is greater than the theoretical value of $F(2, 92, .05) = 3.10$ so that they are statistically significantly different. The geological implications stem from the fact that the pole positions for the two groups are about 10 degrees apart. Two facts suggest that the effects of secular variation have been averaged out. First, several polarity intervals are represented in each group. Second, the angular dispersion of each group is in the order of 11.5 degrees which compares reasonably with Creer's (1962) estimate of about 11 degrees for the same paleolatitude based on the present Earth's magnetic field and summarized paleomagnetic data. Thus it appears that sufficient time must be represented by the flows to allow for several degrees of polar wander. As can be seen in Figure 4a, the wander is perpendicular to the main trend of polar wander during the Tertiary. Therefore the wander is thought to represent a field variation of secondary importance occurring in a time interval spanning less than 10^7 years.

5. Undivided Result

Combining the normalized remanence directions of all the flows giving each unit weight, the resulting mean direction has a cone of 95 per cent confidence of 3.0 degrees and standard error of 1.8 degrees (Fig. 3d, Appendix, Table 3). The geomagnetic pole position derived by assuming a mean sampling site located at 121.8°W, 51.83°N and a geocentric dipole field during extrusion is 140.3°W, 84.0°N with an ellipse of 95 per cent confidence having semiaxes of 4.8 and 5.4 degrees (Fig. 4b). The pole position lies on the polar wandering curve in a location which is consistent with a late Miocene age of extrusion and primary remanence acquisition. The ellipse excludes the Earth's present geomagnetic pole (101.0°W, 75.4°N) which further indicates that the stable primary thermoremanent magnetization (TRM) has been isolated from recently acquired viscous components. The small error and

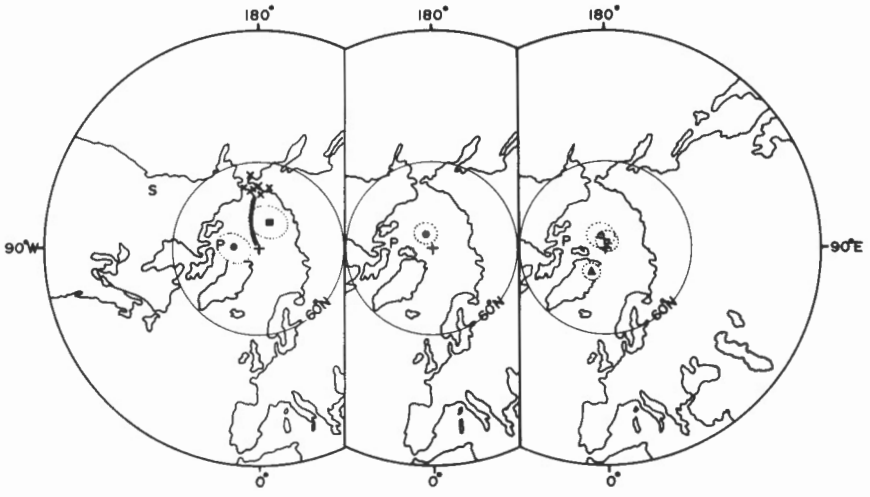


Figure 4. Pole positions shown in the northern hemisphere north of latitude 30° N from the: a) lower (circle) and upper (square) stratigraphic flows with the crosses (X) indicating reliable Cretaceous pole positions from Larochelle (1968) and the heavy line indicating the pole path through the Tertiary; b) all flows (circle); and c) Cariboo gabbroic plugs (circle) (Symons, 1969), the Columbia River basalts (square) and the Oregon plateaus and Owyhee upland (triangle) (Watkins, 1965). The sampling site for the flows is indicated by S, the present geomagnetic pole by P, and the geographic pole by a + cross.

confidence limit values indicate that the flows were deposited during a relatively short geological time interval. It seems likely that about 10 polarity intervals are represented in the sections examined so that a minimum deposition period of 2 x 10^6 years is not unreasonable based on Cox's (op. cit.) estimate. Comparing the thicknesses of the sampled sections with the maximum thickness estimates for the plateau basalts, it seems reasonable to think that no more than 25 polarity intervals are present giving a maximum deposition period of about 5 x 10^6 years. These estimates are consistent with two other lines of evidence. First, the three available age dates for the lower portion of the plateau basalt sequence fall in the 4 x 10^6 year period (of 10 to 13 m.y.) which corresponds closely to the late Miocene age range of 10 to 14 m.y. Additional radiometric age dates on the plateau basalts will help further to delineate the deposition period. Second, sufficient time must be allowed since deposition of the flows for the flows in the extreme western portion of the area to be uplifted some 5,000 feet and dissected an equivalent amount, and for major rivers such as the Fraser and Chilcotin to dissect the plateau basalts by some 2,500 feet. It seems reasonable that most or all of the Pliocene must have been devoted to this process.

6. Comparison with other Paleomagnetic Results

The pole position and ellipse of 95 per cent confidence derived for the plateau basalts is in almost perfect coincidence with that derived from four gabbroic plugs which intrude the lavas (Symons, in press). This result strongly supports the hypothesis advanced by Farquharson (1963) that the plugs occupy the original volcanic vents from which the plateau basalts were extruded (Fig. 4c, Appendix, Table 3).

Watkins (1965) studied Miocene lavas from the Oregon plateaus and Owyhee upland in Oregon and from the Columbia River basalts in Washington and Idaho. Based on paleontological data, the age of the flows which he studied range from Middle Miocene to Lower Pliocene but are dominantly Upper Miocene. The most extensively sampled unit is the Steens basalt which contains a polarity reversal which has been very accurately dated at 15.1 ± 0.3 m.y. (Baksi et al., 1967) and such other dates as are available fall in the range of 14.5 m.y. to 21.3 m.y. (Everenden and James, 1964). Thus it appears that the lava flows sampled by Watkins include the age range of the Cariboo plateau basalts but they are probably older by a few million years on average. As shown in Figure 4c and Table 3, the mean pole positions that Watkins derived from the two areas differ by about 11 degrees arc. He concluded that post-Miocene rotation in the Idaho orocline as proposed by Carey (1958) and Wise (1963) logically explains the angular difference between the pole positions with the Columbia River basalts representing the 'stable' or nonrotated element. The pole position derived here for the Cariboo Plateau basalts supports his conclusion in that it agrees closely with that derived from the Columbia River basalts but not with that obtained for the Oregon plateaus and Owyhee uplands. This result is consistent with the tectonic pattern originally postulated by Carey (1958) in which statistically significant displacement would not be expected between the Cariboo Plateau basalts and Columbia River basalts.

CONCLUSIONS

It is concluded from this study that the Cariboo Plateau basalts as sampled were extruded over a time interval of 2 to 5 m.y. during the late Miocene (14 to 10 m.y. ago). They were deposited in local individual accumulations which built up rapidly within one polarity interval (average duration $\approx .22$ m.y. (Cox, 1968)). Presumably, a fluid magma extruded from numerous volcanic vents to form the accumulations of flat-lying flows which abutted and overlapped to form the present extensive intermontane plateau. Since deposition, they have not been deformed except in local slump structures, nor have they been subjected to regional tectonic rotation except through uplift on their western margins. Thus the pole position derived from their stable primary TRM accurately reflects the Earth's magnetic field during the late Miocene. Its location is consistent with the poles derived from the comagmatic gabbroic intrusives and the Upper Miocene Columbia River basalts.

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APPENDIX

Tables 1-3

Table 1. Site remanence data.

SITE	N	SI	REJECT	n	R	Ds	Is	δ_s	δ_{se}	θ_s	SITE	N	SI	REJECT	n	R	Ds	Is	δ_s	δ_{se}	θ_s
1	8	25.62		8	7.8126	130.4	-74.6	13.3	5.4	1.8	26	5	16.74	1b	4	3.9515	51.6	65.7	10.3	5.1	5.6
2	5	3.84		5	4.8227	125.7	-11.0	17.1	7.6	2.7	27	6	19.62	1b	5	4.9912	43.0	64.4	3.8	1.7	2.0
3	6	30.76	1b	5	4.9878	154.2	-71.9	4.5	2.0	1.3	28	5	40.68		5	4.9924	41.1	66.3	3.5	1.6	1.5
4	5	36.65	1b	4	3.9969	159.3	-72.8	2.6	1.3	2.6	29	6	20.77	1b	5	4.9774	39.9	65.9	6.1	2.7	1.9
5	6	20.94		6	5.9859	170.4	-71.1	4.3	1.8	4.0	30	5	21.10		5	4.9986	31.2	67.7	1.5	0.7	1.9
6	5	34.56		5	4.9775	177.8	-66.2	6.1	2.7	2.3	31	5	27.63		5	4.9984	315.6	78.7	1.6	0.7	2.3
7	3	14.91		3	2.9925	227.9	-75.1	5.0	2.9	1.8	32	5	25.48		5	4.9948	333.9	80.5	2.9	1.3	2.6
8	4	28.14		4	3.9921	201.5	-70.7	4.2	2.1	2.2	33	6	27.45		6	5.9928	316.4	80.6	3.1	1.3	2.5
9	5	22.16		5	4.9924	176.4	-71.7	3.5	1.6	1.9	34	5	27.57		5	4.9884	328.2	76.4	4.4	2.0	1.6
10	5	57.57		5	4.8192	8.8	60.7	17.2	7.7	1.6	35	5	4.80	1a, 1b	3	2.9785	28.1	79.0	8.4	4.8	7.4
11	5	25.65		5	4.9872	206.9	-64.0	4.6	2.1	1.8	36	5	8.84	1b	4	3.9930	131.8	-60.6	3.9	2.0	3.2
12	5	25.42		5	4.9984	179.3	-71.8	1.6	0.7	2.1	37	5	5.70		5	4.9680	125.9	-61.9	7.2	3.2	2.5
13	6	29.22	1a	5	4.9933	337.2	63.6	3.3	1.5	1.5	38	5	19.54		5	4.9731	127.5	-63.2	6.6	3.0	4.9
14	4	42.36		4	3.9939	355.8	62.9	3.7	1.9	2.6	39	4	10.45		4	3.9913	131.8	-66.1	4.4	2.2	4.8
15	5	18.39	1b	4	3.9823	5.1	62.8	6.2	3.1	3.5	40	5	8.60		5	4.9798	131.2	-62.0	5.8	2.6	3.0
16	6	13.76		6	5.9869	350.9	65.9	4.2	1.7	1.8	41	5	34.37		5	4.9896	185.8	-74.7	4.1	1.8	1.6
17	5	13.93	1a	4	3.9944	336.7	71.9	3.5	1.8	1.3	42	5	98.18		5	4.9946	180.5	-73.2	3.0	1.3	1.2
18	5	13.05		5	4.9972	315.4	73.0	2.1	0.9	2.8	43	5	28.41		5	4.9949	186.2	-75.9	2.9	1.3	1.8
19	5	18.56		5	4.9877	352.4	65.0	4.5	2.0	2.8	44	5	29.38		5	4.9902	193.3	-76.4	4.0	1.8	1.5
20	5	9.64		5	4.9924	336.2	69.8	3.5	1.6	3.4	45	5	26.74	1b	4	3.9946	187.9	-72.2	3.4	1.7	1.4
21	5	22.22		5	4.9964	199.6	-66.5	2.4	1.1	2.3	46	5	20.59		5	4.9982	180.0	-73.3	1.7	0.8	1.8
22	5	23.84		5	4.9693	145.5	-73.9	7.1	3.2	1.9	47	5	28.31		5	4.9891	176.1	-64.7	4.2	1.9	2.2
23	6	12.75		6	5.9947	205.2	-50.8	2.6	1.1	2.8	48	5	4.83	1a, 1b	3	2.9898	165.5	-84.0	5.8	3.3	5.3
24	5	68.50		5	4.9945	0.3	73.0	3.0	1.3	0.9	49	6	5.83		6	5.9612	338.2	78.2	7.1	2.9	8.0
25	4	21.87		4	4.9882	49.9	64.5	4.4	2.2	2.2											

N number of cores collected at site

SI stability index

REJECT rejected cores specified by number and reason: (a) angle between the remanence directions of the two cylinders exceeds 15 degrees, (b) single core direction diverges from the mean remanence direction of the remaining cores by more than 3 x δ_s

n number of cores used to calculate R, D_s, I_s, δ_s, δ_{se}, and θ_s.

R resultant giving each core unit weight

$D_s(I_s)$ declination (inclination) of site mean remanence vector in degrees

δ_s angular standard deviation in degrees

δ_{se} angular standard error in degrees

θ_s mean angular deviation between the remanence directions of the two cylinders of each core in degrees

Table 2. Summary of remanence data

SECTION	P	N	R	D_x	I_x	α_{95}	δ_x	δ_{xe}	STRAT
Deadman Falls	R	1		310.4	74.6				M
Clinton Chasm	R	6	5.9432	0.1	72.8	6.1	8.6	3.5	L-M
Tin Cup Mountain	R	1		356.4	71.7				M
Mount Begbie	N	1		8.8	60.7				M
Forestry Hill	R	2	1.9875	15.5	68.5	11.1	9.0	6.4	M
100 Mile House	N	3	2.9876	352.9	63.5	6.4	6.4	3.7	M-U
Clearwater Lake Rd.	N	5	4.9765	340.2	69.6	4.8	6.2	2.8	M-U
Bridge Lake	R	2	1.9726	357.9	72.1	16.4	13.2	9.3	L
Lac la Hache	R	1		25.2	50.8				U
Dog Creek	N	7	6.9583	38.7	67.4	4.4	6.8	2.6	M
Beaumont Creek	N	5	4.9806	335.3	80.1	4.4	5.6	2.5	L
Bull Canyon	R	5	4.9965	309.6	62.7	1.9	2.4	0.8	U
Hanceville	R	7	6.9833	3.4	73.0	2.8	4.3	1.6	U
Heckman Pass	M	2	1.9974	340.7	81.1	5.1	4.2	3.0	L
GROUP				D_g	I_g		δ_g	δ_{ge}	
Normal flows	N	22	21.5299	4.9	72.4	4.5	12.1	2.6	L-U
Reverse flows	R	26	25.4492	349.4	72.0	4.1	12.0	2.4	L-U
Lower flows	M	27	26.5170	9.8	74.0	3.7	11.0	2.1	L-M
Upper flows	M	21	20.5856	343.2	69.1	4.4	11.7	2.5	M-U
All flows	M	48	46.9391	356.4	72.3	3.0	12.2	1.8	L-U

P remanence direction polarity: normal (N), reversed (R), mixed (M)

N number of sites

D(I) declination (inclination) of mean remanence vector by stratigraphic section (subscript x) and group (subscript g) in degrees

α_{95} radius of the cone of 95 per cent confidence in degrees

$\delta_x(\delta_{xe})$ angular standard deviation (error) in degrees

STRAT stratigraphic location: lower (L), middle (M), upper (U)

Table 3. Pole positions

Geologic Unit	Reference	N	B	R	D	I	α_{95}	δ_a	Long.	Lat.	δ_p	δ_m	k
Cariboo plateau basalts	This paper	251	48	46.939	356.4	72.3	3.0	1.8	140.3° W	84.0° N	4.8	5.4	44.3
Cariboo gabbroic plugs	Symons (1969)	101	17	16.864	356.0	71.4	3.1	1.8	146.7° W	84.9° N	4.8	5.5	117.7
Columbia River basalts	Watkins (1965)*	396	198	182.2	356.3	64.9	2.8	1.6	171.7° E	87.1° N	3.7	4.7	12.4
Oregon plateau basalts	Watkins (1965)*	515	235	219.14	13.3	60.8	2.4	1.4	32.2° W	80.1° N	3.0	3.9	12.6

N(B) number of samples (sites)

R resultant of direction cosines

D(I) declination (inclination) of normalized site mean remanence vectors in degrees

α_{95} radius of the cone of 95 per cent confidence in degrees

δ_a angular standard error in degrees

Long. (Lat.) longitude (latitude) of pole position

δ_p (δ_m) semi-axes of the oval of 95 per cent confidence perpendicular (along) the site-pole great circle

k precision estimate (B-1)/(B-R)

* The slight variation in the statistical parameters reported originally and given here result from the recalculation of the original data according to the definitions given by Larochelle (1968).